Shiyu Hu

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https://huuuuusy.github.io/ Google Scholar ResearchGate

Research Interests

I study **the dynamic states of humans and AI in open environments**. As AI moves from answering single questions to carrying out sustained tasks, from long-horizon agents to video and world models and AI working alongside people, the hard part is no longer a correct answer at one moment but keeping the right state: which target is being followed, how long a description holds, which evidence still counts, and where the person stands. Using human performance and human judgment as reference, I evaluate where this state holds and breaks, model systems that keep it, and apply them to UAV vision, video understanding, and education.

Across Conditions	Whether a capability still holds when task and environment conditions change: user-defined task spaces, condition-level benchmarks with human baselines, capability profiles that predict behavior in new settings, and tracking for nighttime UAV vision
Over Time	Whether a system that answers correctly has kept the right state: target identity after disappearance, descriptions that hold only for a stretch of video, language cues that go stale, and long and streaming video understanding
Between Humans and AI	What counts as correct when people disagree or work with AI: human judgments as reference distributions, human–machine comparison on the same tasks, LLM-based virtual students that behave like learners, and AI in education

Each direction runs from evaluation to modeling to application: evaluate where a state holds and breaks, model systems that keep, update, or leave it open, and test them in real settings.

Work Experience

2024.08–Present **Research Fellow, School of Physical and Mathematical Sciences (SPMS), Nanyang Technological University (NTU)**
- **Research:** Computer Vision, Multimodal AI, and AI for Education
- **PI:** Assoc. Prof. Kang Hao Cheong (IEEE Senior Member)

2018.02–2018.12 **Research Assistant, University of Hong Kong (HKU)**
- **Research:** High Performance Computing, Heterogeneous Computing
- **PI:** Prof. Choli Wang

2016.08–2016.09 **Research Intern, Institute of Electronics, Chinese Academy of Sciences (CASIE)**

Education

2019.09–2024.01 **Ph.D., Institute of Automation, Chinese Academy of Sciences (CASIA)**
- **Field:** Computer Application Technology
- **Supervisor:** Prof. Kaiqi Huang (IAPR Fellow, IEEE Senior Member)
- **Co-supervisor:** Prof. Xin Zhao (IEEE Senior Member)
- **Thesis title:** *Research of Intelligence Evaluation Techniques for Single Object Tracking*
- **Thesis committee:** Profs. Jianbin Jiao, Yuxin Peng (CAAI/CIE/CSIG Fellow), Yao Zhao (IEEE/IET Fellow), Yunhong Wang (IEEE/IAPR/CCF Fellow), and Ming Tang
- **Thesis defense grade:** Excellent

2017.09–2019.07 **M.Sc., Department of Computer Science, University of Hong Kong (HKU)**
- **Field:** Computer Science
- **Supervisor:** Prof. Choli Wang
- **Thesis title:** *NightRunner: Deep Learning for Autonomous Driving Cars after Dark*
- **Thesis defense grade:** A+

Education (continued)

2013.09–2017.07 **B.E., Elite Class in School of Information and Electronics, Beijing Institute of Technology (BIT)**
- **Field:** Information Engineering
- **Supervisor:** Prof. Senlin Luo
- **Thesis title:** *Text Sentiment Analysis Based on Deep Neural Network*
- **Thesis defense grade:** Excellent

2015.07–2015.08 **Summer Session, University of California, Berkeley (UC Berkeley)**
- **Course:** New Media
- **Course grade:** A

Research Projects and Platforms

Funded Research Projects

2025–Present **Singapore MOE Tertiary Education Research Fund** (MOE2024-TRF-004)
- **Core Researcher**, Nanyang Technological University (NTU Singapore)

2023–2025 **Singapore MOE Tertiary Education Research Fund** (MOE2022-TRF-029)
- **Core Researcher**, Nanyang Technological University (NTU Singapore)

2024–2025 **2023 Intelligent Education PhD Research Fund**
- **Research Contributor**, Shanghai Institute of AI Education, ECNU

Research Platforms

2019.11–Present **VideoCube / MGIT Platform** http://videocube.aitestunion.com
- **Role:** builder and maintainer; evaluation infrastructure for the TPAMI 2023 and NeurIPS 2023 studies
- **Impact (July 2026):** 1.66M+ visits, 1.9k+ data users, 720+ registered trackers, 960+ result submissions

2021.07–Present **SOTVerse / VLTVerse Platform** http://metaverse.aitestunion.com
- **Role:** builder and maintainer; task-space evaluation platform for the IJCV 2024 study
- **Impact (July 2026):** 267k+ visits

2022.05–Present **BioDrone Platform** http://biodrone.aitestunion.com/
- **Role:** builder and maintainer; benchmark platform for real-world drone tracking (IJCV 2024)
- **Impact (July 2026):** 541k+ visits

2020.07–Present **GOT-10k Platform** http://got-10k.aitestunion.com/
- **Role:** maintainer; evaluation platform for the TPAMI 2021 study
- **Impact (July 2026):** 7.56M+ visits, 11k+ data users, 32.7k+ registered trackers, 406k+ result submissions

Challenges

2023.05–2023.11 **Hislopvision Challenge** http://hislopvision.aitestunion.com/
- Organized the Hislopvision track of the 3rd High-speed and Low-power Visual Understanding Challenge at PRCV 2023; participating teams from Tsinghua University, Beijing Institute of Technology, and Jilin University

2021.01–2021.04 **Cell Tracking Challenge** https://celltrackingchallenge.net/
- 2nd place on Fluo-C2FL-MSC+ and 3rd place on Fluo-C2FL-Huh7 (rankings as of Oct. 2023)

Research Software

2018.02–2018.12 **Darknet-Cross** https://github.com/huuuuusy/Darknet-Cross
- Lightweight deep learning framework for heterogeneous computing: cross-platform acceleration for Android and Ubuntu across mobile and desktop GPUs; HKU master's thesis project

Academic Activities and Services

Tutorials

Organizer and speaker of all tutorials listed below.

2025.10.26 **28th European Conference on Artificial Intelligence (ECAI 2025, CCF-B)**, Bologna, Italy
Title: From Benchmarking to Trustworthy AI: Rethinking Evaluation Methods Across Vision and Complex Systems

2025.10.05 **IEEE International Conference on Systems, Man, and Cybernetics (SMC 2025, CCF-C)**, Vienna, Austria
Title: The Synergy of Large Language Models and Evolutionary Optimization on Complex Networks

2025.08.18 **34th International Joint Conference on Artificial Intelligence (IJCAI 2025, CCF-B)**, Montreal, Canada
Title: Human-Centric and Multimodal Evaluation for Explainable AI: Moving Beyond Benchmarks

2024.12.09 **17th Asian Conference on Computer Vision (ACCV 2024, CCF-C)**, Hanoi, Vietnam
Title: From Machine-Machine Comparison to Human-Machine Comparison: Adapting Visual Turing Test in Visual Object Tracking

2024.12.01 **27th International Conference on Pattern Recognition (ICPR 2024, CCF-C)**, Kolkata, India
Title: Visual Turing Test in Visual Object Tracking: A New Vision Intelligence Evaluation Technique based on Human-Machine Comparison

2024.10.27 **31st IEEE International Conference on Image Processing (ICIP 2024, CCF-C)**, Abu Dhabi, United Arab Emirates
Title: An Evaluation Perspective in Visual Object Tracking: from Task Design to Benchmark Construction and Algorithm Analysis

Talks and Mini-Symposia

2026.11 **Talk, College of Liberal Arts, Wenzhou-Kean University**, Wenzhou, China
Series: CLA Invited Academic Talks
Title: Studying the Human Mind with AI: From Behavioral Analysis to Role Simulation

2026.10.04 **Talk, 14th International Conference on Image and Graphics (ICIG 2026, CCF-C)**, Singapore
Forum: Visual Intelligence in Transition: From Physical Perception to Psychological Cognition
Title: From Object Tracking to Process Understanding: State Modeling in Dynamic Vision

2026.09.20 **Mini-Symposium, 5th International Nonlinear Dynamics Conference (NODYCON 2026)**, Rome, Italy
Title: Complex Network Systems and Large Language Models

2026.05.29 **Talk, Chinese Congress on Image and Graphics (CCIG 2026)**, Guangzhou, China
Forum: Intelligent Evolution of Video and Image Security: Perception, Reasoning, and Adversarial Challenges
Title: Visual Understanding Reliability in Open Environments: from Robust Perception to Semantic Consistency

2025.12.13 **Alumni Sharing, UCAS Graduate Academic Forum (AI Sub-forum)**, Beijing, China

2025.08.30 **Talk, Visual Intelligence International Postdoctoral Frontier Seminar**, Online
Title: Human-Centric Multimodal Visual Intelligence: From Benchmarks to Human-Like Modeling

Conference Organization and Program Committees

Area Chair **15th International Conference on Learning Representations (ICLR, CCF-A)**

TPC Member **34th Pacific Conference on Computer Graphics and Applications (PG 2026, CCF-B)**

Academic Activities and Services (continued)

Publicity Chair — **CSIG Annual Conference on Video and Image Security 2026**
Date: 2026.11.21
Location: Xiong'an, China

Editorial Roles

Collection Host — **The Innovation Informatics**, Virtual Collection: Control and Decision Intelligence for Unmanned Systems (submissions until 2027.06.30)

Lead Guest Editor — **Electronics**, Special Issues: Techniques and Applications of Multimodal Data Fusion – 1st and 2nd Editions

Associate Editor — **Innovation and Emerging Technologies**

Peer Review

Conferences — NeurIPS, ICML, ICLR, CVPR, ECCV, ICCV, ACL, AAAI, IJCAI, ACM MM, ICRA, and AISTATS

Journals — ACM Computing Surveys; IEEE Transactions on Image Processing; SCIENCE CHINA Information Sciences; Pattern Recognition; Transactions on Machine Learning Research; IEEE Transactions on Network Science and Engineering; IEEE Transactions on Vehicular Technology; Information Fusion; Visual Intelligence; Engineering Applications of Artificial Intelligence; Expert Systems with Applications; Neurocomputing; Knowledge-Based Systems

Professional Memberships

Member — **Committee:** Technical Committee on Video and Image Security (CSIG-TCVIS)
Societies: Institute of Electrical and Electronics Engineers (IEEE); Chinese Association for Artificial Intelligence (CAAI); and China Computer Federation (CCF)

Student Mentoring and Co-Supervision

Joint publications with mentees include papers at CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, ACL, EMNLP, AAAI, IJCAI, and ICRA, and in IJCV, TCSVT, and Pattern Recognition; ongoing mentoring is listed first.

Ph.D. Students

2023.08–Present — **Yiping Ma**, East China Normal University
2023.08–Present — **Dailing Zhang**, Institute of Automation, Chinese Academy of Sciences
2024.08–Present — **Xuchen Li**, Institute of Automation, Chinese Academy of Sciences
2024.08–Present — **Yipei Wang**, Southeast University
2025.01–Present — **Shukun Jia**, Southeast University
2025.01–Present — **Buyuan Zhu**, Nanyang Technological University, Singapore
2025.08–Present — **Zi Ye**, University of Science and Technology Beijing
2025.08–Present — **Yihan Meng**, University of Science and Technology Beijing
2026.02–Present — **Shunya Hirashima**, Nanyang Technological University, Singapore
2022.08–2025.06 — **Meiqi Wu**, University of Chinese Academy of Sciences
2023.04–2026.06 — **Xiaokun Feng**, Institute of Automation, Chinese Academy of Sciences

M.S. Students

2024.07–Present — **Xuzhao Li**, Beijing Institute of Technology
2024.09–Present — **Panxi Xu**, University of Science and Technology Beijing
2025.07–Present — **Hongfei Ma**, University of Science and Technology Beijing
2022.05–2023.07 — **Yiping Ma**, Nanjing Normal University
2022.08–2024.07 — **Yipei Wang**, Southeast University
2024.07–2025.01 — **Yuqi Cui**, University of Science and Technology Beijing
2025.02–2025.11 — **Nguyen Khanh Truong**, Nanyang Technological University, Singapore

Student Mentoring and Co-Supervision (continued)

B.E. Students

2022.09–2023.08 **Junyou Zhu**, University of Chinese Academy of Sciences
2022.09–2023.08 **Lihang Hu**, University of Chinese Academy of Sciences
2022.09–2023.08 **Dailing Zhang**, Southeast University
2023.04–2024.07 **Xuchen Li**, Beijing University of Posts and Telecommunications
2024.09–2025.05 **Le Ying Tan**, Nanyang Technological University, Singapore
2024.09–2025.07 **Zi Ye**, University of Science and Technology Beijing
2024.10–2025.07 **Jinlin Ma**, Nanyang Technological University, Singapore
2024.10–2026.05 **Yuxiao Li**, Nanyang Technological University, Singapore
2024.12–2025.07 **Yihan Meng**, University of Science and Technology Beijing
2025.01–2025.05 **Yihui Jiang**, University of Science and Technology Beijing
2025.04–2025.07 **Shunya Hirashima**, Nanyang Technological University, Singapore
2025.07–2025.09 **Anastasiia Bohachenko**, Nanyang Technological University, Singapore (Global Connect Fellow)
2025.08–2026.05 **Jia Ling Ow**, Nanyang Technological University, Singapore
2025.08–2026.05 **Jian Feng Ho**, Nanyang Technological University, Singapore

Honors and Awards

Professional and Graduate Honors

2026 **Reviewer Award**, 43rd International Conference on Machine Learning (ICML 2026)
2025 **IEEE SMCS TEAM Program Award**, IEEE Systems, Man, and Cybernetics Society
2024 **Best Paper Honorable Mention**, 3rd Workshop on Vision Datasets Understanding and DataCV Challenge in CVPR 2024
Beijing Outstanding Graduate, Beijing Municipal Education Commission (Top 5%)
2023 **China National Scholarship**, Ministry of Education of the People's Republic of China (Top 1%)
First Prize of Climbing Scholarship, Institute of Automation, Chinese Academy of Sciences
2022 **Merit Student**, University of Chinese Academy of Sciences

Undergraduate Honors

Competitions **National First Prize**, Contemporary Undergraduate Mathematical Contest in Modeling (2015, Top 1%); **First Prize**, Mathematical Modeling Competition, Beijing Institute of Technology (2015)
Scholarships **Academic Scholarship**, Beijing Institute of Technology (2013–2017); **College Scholarship**, Chinese Academy of Sciences (2016)
Student Honors **Excellent Innovative Student** (2017), **Outstanding Student Cadre** (2015), and **Outstanding Student** (2014), Beijing Institute of Technology
Leadership & Service Youth Day Competition, Beijing Institute of Technology: **Excellent League Member** (2016), **Outstanding League Cadre** (2015), **Outstanding Youth League Branch** (2015, Team Leader), and **Top 10 Activities** (2015, Team Leader); Summer Social Practice: **Outstanding Individual** (2015) and **Second Prize** (2015, Team Leader)

Skills & Interests

Programming Python, MATLAB, C, Java, JavaScript, HTML/CSS, Shell, Git, and LaTeX; LLM-assisted coding
Systems Research-platform and Flask/SQLite web development; Linux, Android, and heterogeneous computing

Skills & Interests (continued)

Leadership	Research supervision; grant writing; project delivery; cross-institutional collaboration; academic writing, peer review, tutorials, and presentations
Languages	Chinese (native) and English
Hobbies	Art, writing, LEGO, fitness, and former amateur marathoner

Research Publications

Publications: 1 monograph; 42 journal and conference papers (14 in CCF-A venues; 9 as first or corresponding author: TPAMI, IJCV ×2, NeurIPS, ICCV Highlights, ICRA, ICMR, IET-CVI, and a JIG survey); 4 workshop papers; 11 preprints
Citations: Google Scholar, Oct. 2026: 1,724 citations, h-index 17, i10-index 26
Notation: ∗ equal contribution; ✉ corresponding author; CCF-T1/T2/T3: CCF 2025 tiers for Chinese journals; JCR-indexed journals: JCR quartile and 2025 IF
Icons: paper; survey or perspective; monograph; a dashed frame marks a preprint

Monograph

[B1] X. Zhao, **S. Hu**, and X. Yin, *Visual Object Tracking: An Evaluation Perspective*. Springer, 2025, ISBN: 978-981-96-4558-9. DOI: 10.1007/978-981-96-4558-9

First- and Corresponding-Author Publications

[A1] **S. Hu**, X. Zhao, L. Huang, and K. Huang, "Global instance tracking: Locating target more like humans," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **(TPAMI, CCF-A, JCR Q1, IF 20.4)**, vol. 45, no. 1, pp. 576–592, 2023. DOI: 10.1109/TPAMI.2022.3153312

[A2] **S. Hu**, X. Zhao, and K. Huang, "SOTVerse: A user-defined task space of single object tracking," *International Journal of Computer Vision* **(IJCV, CCF-A, JCR Q1, IF 10.3)**, vol. 132, pp. 872–930, 2024. DOI: 10.1007/s11263-023-01908-5

[A3] X. Zhao, **S. Hu**✉, Y. Wang, J. Zhang, Y. Hu, R. Liu, H. Ling, Y. Li, R. Li, K. Liu, and J. Li, "BioDrone: A bionic drone-based single object tracking benchmark for robust vision," *International Journal of Computer Vision* **(IJCV, CCF-A, JCR Q1, IF 10.3)**, vol. 132, pp. 1659–1684, 2024. DOI: 10.1007/s11263-023-01937-0

[A4] **S. Hu**, D. Zhang, M. Wu, X. Feng, X. Li, X. Zhao, and K. Huang, "A multi-modal global instance tracking benchmark (MGIT): Better locating target in complex spatio-temporal and causal relationship," *Advances in Neural Information Processing Systems* **(NeurIPS, CCF-A)**, vol. 36, pp. 25 007–25 030, 2023.

[A5] X. Feng∗, **S. Hu**∗, X. Li, D. Zhang, M. Wu, J. Zhang, X. Chen, and K. Huang, "ATCTrack: Aligning target-context cues with dynamic target states for robust vision-language tracking," in *Proceedings of the IEEE/CVF International Conference on Computer Vision* (**ICCV, CCF-A**; **Highlights**), 2025, pp. 19 850–19 861.

[A6] X. Li∗, X. Li∗, and **S. Hu**✉, "MATrack: Efficient multiscale adaptive tracker for real-time nighttime UAV operations," in *Proceedings of the IEEE International Conference on Robotics and Automation* **(ICRA, CCF-B)**, 2026.

[A7] X. Li∗, X. Li∗, and **S. Hu**✉, "DARTer: Dynamic adaptive representation tracker for nighttime UAV tracking," in *Proceedings of the International Conference on Multimedia Retrieval* **(ICMR, CCF-B)**, 2025.

[A8] **S. Hu**, X. Zhao, and K. Huang, "Visual intelligence evaluation techniques for single object tracking: A survey," *Journal of Image and Graphics* (《中国图象图形学报》) **(CCF-T2)**, vol. 29, no. 8, pp. 2269–2302, 2024. DOI: 10.11834/jig.230498

[A9] Y. Wang, J. Zhang, Y. Wang, **S. Hu**✉, B. Shen, Z. Hou, and W. Zhou, "Improved SAR aircraft detection algorithm based on visual state space models," *IET Computer Vision* **(IET-CVI, CCF-C, JCR Q4, IF 1.4)**, vol. 19, no. 1, e70032, 2025.

Collaborative Publications

[C10] P. Zhao, **S. Hu**, J. Zhang, and M. Li, "META-PAP: Meta-learning for prompt-aware preference pairing in LLM alignment," *Advances in Neural Information Processing Systems* **(NeurIPS, CCF-A)**, 2026.

[C11] X. Li, X. Li, **S. Hu**, and K. Huang, "Select less, reason more: Prioritizing evidence purity for video reasoning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* **(CVPR, CCF-A)**, 2026.

[C12] X. Li, X. Li, **S. Hu**, K. Huang, and W. Zhang, "CausalStep: A benchmark for explicit stepwise causal reasoning in videos," in *Proceedings of the AAAI Conference on Artificial Intelligence* (**AAAI, CCF-A**; **Oral**), 2026.

[C13] X. Li, X. Li, **S. Hu**, Y. Guo, and W. Zhang, "VerifyBench: A systematic benchmark for evaluating reasoning verifiers across domains," in *Proceedings of the AAAI Conference on Artificial Intelligence* (**AAAI, CCF-A**; **Oral**), 2026.

[C14] X. Feng, H. Yu, M. Wu, **S. Hu**, J. Chen, C. Zhu, J. Wu, X. Chu, and K. Huang, "NarrLV: Towards a comprehensive narrative-centric evaluation for long video generation models," in *Proceedings of the International Conference on Learning Representations* **(ICLR, CCF-A)**, 2026.

[C15] X. Li, X. Li, J. Gao, R. Pi, **S. Hu**, and W. Zhang, "Look less, reason more: Rollout-guided adaptive pixel-space reasoning," in *Proceedings of the Annual Meeting of the Association for Computational Linguistics* **(ACL, CCF-A)**, 2026.

[C16] S. Jia, **S. Hu**, Y. Cao, F. Yang, X. Lu, and X. Lu, "Tracking by detection and query: An efficient end-to-end framework for multi-object tracking," *Pattern Recognition* **(PR, CCF-B, JCR Q1, IF 9.1)**, 2026.

[C17] M. Wu, Y. Zhao, X. Li, **S. Hu**, Y. Cai, J. Wu, W. Wang, and K. Huang, "Talk with your fingers: A depth-aware benchmark for air-writing recognition," *IEEE Transactions on Circuits and Systems for Video Technology* **(TCSVT, CCF-B, JCR Q1, IF 10.8)**, 2026. DOI: 10.1109/TCSVT.2026.3670256

[C18] S. Jia, **S. Hu**, Y. Wang, X. Cheng, Y. Cao, and X. Lu, "COAL: Counterfactual and observation-enhanced alignment learning for discriminative referring multi-object tracking," in *the 35th International Joint Conference on Artificial Intelligence* **(IJCAI, CCF-B)**, 2026.

[C19] D. Zhang, **S. Hu**, H. Fu, X. Feng, Y. Wang, K. H. Cheong, and K. Huang, "Rethinking temporal modeling in visual object tracking via decoupled auxiliary supervision," in *Proceedings of the European Conference on Computer Vision* **(ECCV, CCF-B)**, 2026.

[C20] M. Peng, H. Gu, Y. Ma, **S. Hu**, and X. Huang, "Fake news court: A multi-agent adversarial framework for robust detection of LLM-generated fake news," in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing* **(EMNLP, CCF-B)**, 2026.

[C21] X. Li, X. Li, R. Pi, **S. Hu**, J. Zhao, and J. Gao, "Beyond accuracy: Evaluating grounded visual evidence in thinking with images," in *Findings of the Association for Computational Linguistics: EMNLP 2026* **(EMNLP Findings)**, 2026.

[C22] K. Li, B. Parikh, H. Yue, **S. Hu**, S. W. Tan, W. Y. Low, X. Su, and K. H. Cheong, "Global-local semi-supervised modeling for retinal layer boundary estimation in OCT," in *the 48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society* **(EMBC, CAAI-B)**, 2026.

[C23] X. Li, X. Li, **S. Hu**, Z. Zhang, and K. H. Cheong, "Constraint-driven evolution of multimodal video intelligence: A network and system perspective," *IEEE Transactions on Network Science and Engineering* **(TNSE, JCR Q1, IF 7.3)**, 2026.

[C24] S. Hirashima, **S. Hu**, and K. H. Cheong, "Beyond hyper-rationality: Persona-driven heterogeneity in networked agent-based systems," *IEEE Transactions on Network Science and Engineering* **(TNSE, JCR Q1, IF 7.3)**, 2026.

[C25] Y. Ma, M. Peng, **S. Hu**, B. Zhu, Y. Wang, Y. Chen, S. Liu, and K. H. Cheong, "From observation to generation: A complex-systems perspective on generative human-AI social systems," *Chaos, Solitons & Fractals* **(CSF, JCR Q1, IF 5.7)**, 2026.

[C26] L. Tan, B. Zhu, **S. Hu**, A. Mishra, D. J. Yeo, and K. H. Cheong, "CalcTutor: Multi-agent LLM grading of handwritten mathematics with RAG-grounded feedback for adaptive learning support," *Mathematics* **(JCR Q1, IF 2.3)**, 2026.

[C27] Center for AI Safety, Scale AI, and HLE Contributors Consortium (including **Shiyu Hu**), "A benchmark of expert-level academic questions to assess AI capabilities," *Nature* **(JCR Q1, IF 56.1)**, vol. 649, pp. 1139–1146, 2026. DOI: 10.1038/s41586-025-09962-4

[C28] X. Feng, D. Zhang, **S. Hu**, X. Li, M. Wu, J. Zhang, X. Chen, and K. Huang, "CSTrack: Enhancing RGB-X tracking via compact spatiotemporal features," in *Proceedings of the International Conference on Machine Learning* **(ICML, CCF-A)**, 2025.

[C29] X. Feng, D. Zhang, **S. Hu**, X. Li, M. Wu, J. Zhang, X. Chen, and K. Huang, "Enhancing vision-language tracking by effectively converting textual cues into visual cues," in *IEEE International Conference on Acoustics, Speech, and Signal Processing* **(ICASSP, CCF-B)**, 2025.

[C30] L. Tan, **S. Hu**, D. J. Yeo, and K. H. Cheong, "Artificial intelligence-enabled adaptive learning platforms: A review," *Computers & Education: Artificial Intelligence* **(C&E:AI, JCR Q1, IF 23.4)** *(No. 3 of 210 in Artificial Intelligence; highest JIF among AI-in-education journals)*, p. 100 429, 2025.

[C31] L. Tan, **S. Hu**, D. J. Yeo, and K. H. Cheong, "A comprehensive review on automated grading systems in STEM using AI techniques," *Mathematics* **(JCR Q1, IF 2.3)**, vol. 13, no. 17, 2025.

[C32] Y. Ma, X. Li, **S. Hu**, S. Liu, and K. H. Cheong, "Trustworthy AI in education: Framework, cases, and governance strategies," *Innovation and Emerging Technologies* **(JCR Q2, IF 2.3)**, vol. 12, p. 2 550 026, 2025.

[C33] K. Huang, Y. Kang, C. Yan, **S. Hu**, L. Wang, T. Tao, and W. Gao, "A review of intelligent psychological assessment based on interactive environment," *Chinese Mental Health Journal* (《中国心理卫生杂志》) **(CSCD Core, PKU Core)**, 2025.

[C34] D. Zhang, **S. Hu**, X. Feng, X. Li, M. Wu, J. Zhang, and K. Huang, "Beyond accuracy: Tracking more like human via visual search," *Advances in Neural Information Processing Systems* **(NeurIPS, CCF-A)**, vol. 37, pp. 2629–2662, 2024.

[C35] X. Feng, X. Li, **S. Hu**, D. Zhang, M. Wu, J. Zhang, X. Chen, and K. Huang, "MemVLT: Visual-language tracking with adaptive memory-based prompts," *Advances in Neural Information Processing Systems* **(NeurIPS, CCF-A)**, vol. 37, pp. 14 903–14 933, 2024.

[C36] J. Zhang, T. Zhao, **S. Hu**, and X. Zhao, "Robust single-particle Cryo-EM image denoising and restoration," in *IEEE International Conference on Acoustics, Speech, and Signal Processing* **(ICASSP, CCF-B)**, 2024.

[C37] M. Wu, K. Huang, Y. Cai, **S. Hu**, Y. Zhao, and W. Wang, "Finger in camera speaks everything: Unconstrained air-writing for real-world," *IEEE Transactions on Circuits and Systems for Video Technology* **(TCSVT, CCF-B, JCR Q1, IF 10.8)**, vol. 34, no. 9, pp. 8602–8613, 2024.

[C38] M. Wu, Y. Kang, X. Li, **S. Hu**, X. Chen, Y. Kang, W. Wang, and K. Huang, "VS-LLM: Visual-semantic depression assessment based on LLM for drawing projection test," in *Chinese Conference on Pattern Recognition and Computer Vision* **(PRCV, CCF-C)**, 2024, pp. 232–246.

[C39] X. Feng, **S. Hu**, X. Chen, and K. Huang, "A hierarchical theme recognition model for sandplay therapy," in *Chinese Conference on Pattern Recognition and Computer Vision* **(PRCV, CCF-C)**, 2023, pp. 241–252.

[C40] Y. Wang, **S. Hu**, and X. Zhao, "Rethinking similar object interference in single object tracking," in *International Conference on Computer Science and Artificial Intelligence* (**CSAI, EI**; **Oral**), 2023, pp. 251–258.

[C41] Y. Zhang, C. Liu, W. Chen, X. Xu, F. Wang, H. Li, **S. Hu**, and X. Zhao, "Revisiting instance search: A new benchmark using cycle self-training," *Neurocomputing* **(Neu, CCF-C, JCR Q1, IF 6.7)**, vol. 501, pp. 270–284, 2022. DOI: 10.1016/j.neucom.2022.06.027

[C42] K. Huang, X. Zhao, Q. Li, and **S. Hu**, "Visual turing: The next development of computer vision in the view of human-computer gaming," *Journal of Graphics* (《图学学报》) **(CCF-T3)**, vol. 42, no. 3, p. 339, 2021. DOI: 10.11996/JG.j.2095-302X.2021030339

Workshop Papers

[W1] Y. Ma∗, **S. Hu**∗, X. Li, Y. Wang, Y. Chen, S. Liu, and K. H. Cheong, "Learning to be taught: A structured SOEI framework for modeling and evaluating personality-aligned virtual student agents," in *AAAI Conference on Artificial Intelligence AI4Edu Workshop* (**AAAIW, CCF-A Workshop**; **Oral**), 2026.

[W2] Y. Ma∗, **S. Hu**∗, B. Zhu, Y. Wang, Y. Kang, S. Liu, and K. H. Cheong, "Redefining educational simulation: EduVerse as a user-defined and developmental multi-agent simulation space," in *AAAI Conference on Artificial Intelligence AI4Edu Workshop* (**AAAIW, CCF-A Workshop**; **Oral**), 2026.

[W3] B. Zhu∗, **S. Hu**∗, Y. Ma, Y. Zhang, and K. H. Cheong, "From objective to subjective: A benchmark for virtual student abilities," in *AAAI Conference on Artificial Intelligence AI4Edu Workshop* (**AAAIW, CCF-A Workshop**; **Oral**), 2026.

[W4] X. Li, X. Feng, **S. Hu**, M. Wu, D. Zhang, J. Zhang, and K. Huang, "DTLLM-VLT: Diverse text generation for visual language tracking based on LLM," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition 3rd VDU Workshop* (**CVPRW, CCF-A Workshop**; **Oral, Best Paper Honorable Mention**), 2024.

Preprints

[P1] X. Li∗, **S. Hu**∗, X. Feng, J. Zhao, and K. Huang, "How to respond, how to memorize, how to be fast: A survey of streaming video understanding," *SSRN preprint*, 2026. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7008298

[P2] **S. Hu**∗, X. Li∗, X. Li, J. Zhang, Y. Wang, X. Zhao, and K. H. Cheong, "FIOVA: A multi-annotator benchmark for human-aligned video captioning," *arXiv preprint arXiv:2410.15270*, 2024.

[P3] Y. Ma∗, **S. Hu**∗, X. Li, Y. Wang, Y. Chen, S. Liu, and K. H. Cheong, "When LLMs learn to be students: The SOEI framework for modeling and evaluating virtual student agents in educational interaction," *arXiv preprint arXiv:2410.15701*, 2024.

[P4] Y. Ma∗, **S. Hu**∗, B. Zhu, Y. Wang, Y. Kang, S. Liu, and K. H. Cheong, "EduVerse: A user-defined multi-agent simulation space for education scenario," *arXiv preprint arXiv:2510.05650*, 2025.

[P5] B. Zhu∗, **S. Hu**∗, Y. Ma, Y. Zhang, and K. H. Cheong, "EduPersona: Benchmarking subjective ability boundaries of virtual student agents," *arXiv preprint arXiv:2510.04648*, 2025.

[P6] Y. Wang∗, **S. Hu**∗, S. Jia, P. Xu, H. Ma, Y. Ma, J. Zhang, X. Lu, and X. Zhao, "SOI is the root of all evil: Quantifying and breaking similar object interference in single object tracking," *arXiv preprint arXiv:2508.09524*, 2025.

[P7] X. Li∗, **S. Hu**∗, X. Feng, D. Zhang, M. Wu, J. Zhang, and K. Huang, "How texts help? a fine-grained evaluation to reveal the role of language in vision-language tracking," *arXiv preprint arXiv:2411.15600*, 2024.

[P8] X. Li, X. Li, J. Zhao, and **S. Hu**✉, "STEMVerse: A dual-axis diagnostic framework for STEM reasoning in large language models," *arXiv preprint arXiv:2602.02497*, 2026.

[P9] X. Li, **S. Hu**, X. Feng, D. Zhang, M. Wu, J. Zhang, and K. Huang, "DTVLT: A multi-modal diverse text benchmark for visual language tracking based on LLM," *arXiv preprint arXiv:2410.02492*, 2024.

[P10] X. Li, **S. Hu**, X. Feng, D. Zhang, M. Wu, J. Zhang, and K. Huang, "Visual language tracking with multi-modal interaction: A robust benchmark," *arXiv preprint arXiv:2409.08887*, 2024.

[P11] **S. Hu**, X. Zhao, Y. Wang, Y. Shan, and K. Huang, *Nearing or surpassing: Overall evaluation of human-machine dynamic vision ability*, 2023. URL: https://openreview.net/forum?id=LGbzYw_pnsc

References

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